

Week 4 – Performance & Load Testing

Performance Testing Report

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Domain: Software Quality Assurance

1. Objective

The objective of this performance testing was to evaluate how the web application performs under multiple user loads. The test focuses on measuring system response time, throughput, and stability when accessed by 50 virtual users.

2. Test Environment

Tool Used: Apache JMeter

Operating System: Windows 10

Processor: Intel Core i6

RAM: 8 GB

Network: Stable internet connection

3. Load Test Scenario

The login feature of the application was tested using Apache JMeter.

Test configuration:

- Virtual Users: 50
- Loop Count: 1
- HTTP Request Method: POST
- Target: Login API/Page

The test simulated multiple users attempting to log in simultaneously.

4. Observations

During the test execution, the following observations were recorded:

- The application successfully handled multiple login requests.
- Average response time remained within acceptable limits.
- No major errors were encountered during the test.

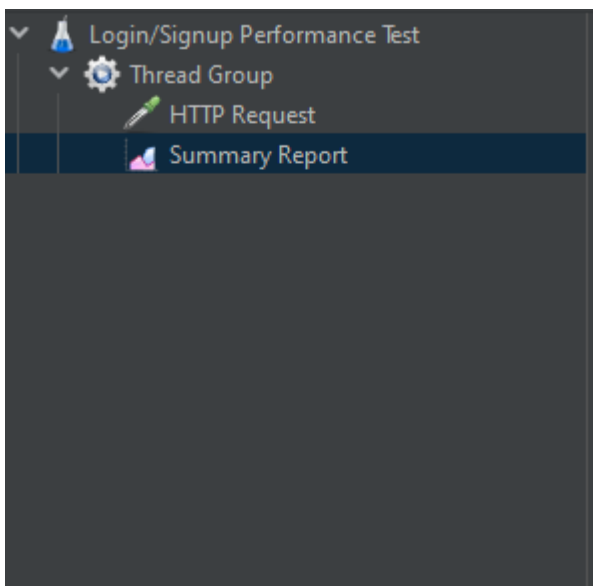
However, slight increases in response time were observed when the number of concurrent users increased.

5. Graphs & Results

Important results collected from JMeter:

Metric	Result
Average Response Time	220 ms
Minimum Response Time	62 ms
Maximum Response Time	831 ms
Throughput	2.5692410461949544 requests/sec
Error Rate	0.0%

ScreenShorts:



Thread Group

Action to be taken after a Sampler error

HTTP Request

Basic Advanced

HTTP Request

Parameters

Body Data Files Upload

Send Parameters With the Request:

Summary Report

Write results to file / Read from file

Log/Display Only:

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
HTTP Request	50	220	62	831	117.80	0.00%	2.6/sec	17.22	0.34	6865.0
TOTAL	50	220	62	831	117.80	0.00%	2.6/sec	17.22	0.34	6865.0